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Hybrid AI Framework for Dyslexia Detection Using MFCC Features, Optimized SVM, and Reinforcement-Based Assistive Learning

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***Abstract*—A pervasive neurodevelopmental disorder, dyslexia has a substantial influence on reading, writing, and cognitive processing, which impacts everyday living as well as academic achievement. This work offers a hybrid AI-driven method for detecting dyslexia that combines adaptive reinforcement learning, speech signal analysis, and EEG-based biomarkers to improve diagnosis accuracy and offer individualized assistance. To enhance classification performance, the suggested system uses optimal Support Vector Machine (SVM) classifiers with sophisticated preprocessing methods, such as feature extraction and noise reduction. Real-time processing, scalability, and accessibility are guaranteed by a Raspberry Pi-based design, which makes it appropriate for a range of clinical and educational applications. The system's capacity to differentiate between dyslexic speech patterns and neurophysiological signals is confirmed by experimental evaluations, guaranteeing precise early diagnosis. Additionally, by tailoring interventions to each student's requirements, the adaptive reinforcement learning module increases engagement and enhances learning results. The results demonstrate the approach's promise as a thorough, economical, and successful method of dyslexia screening and assistance.**

Keywords: Dyslexia diagnosis, Advanced machine learning, Edge computing with Raspberry Pi, SVM classifiers, Adaptive reinforcement learning

I. Introduction

Millions of people worldwide suffer with dyslexia, a neurodevelopmental disease that causes major difficulties with phonological processing, reading, and writing, particularly throughout the school years. Even though our knowledge of its neurological and cognitive basis has significantly improved, early identification and successful treatment are still quite important. Access is restricted due to the time-consuming, expensive, and laborious nature of traditional diagnostic techniques, mainly in underprivileged areas. Through a combination of AI and ML, the diagnosis of dyslexia can be transformed into an automated, real-time, and scalable solution. Speech processing has emerged as a strong diagnostic tool for dyslexia, especially when combined with neurophysiological data analysis, such as EEG. The multimodal approach in this study offers a very comprehensive approach to the exploration of dyslexia using behavioral data, neurophysiological measures, and speech data together. Some possible future enhancement may include using cloud computing for data management and easy access to integrating IoT devices. This paper contributes to the AI-neurobiology literature by providing useful solutions on the diagnosis and treatment of dyslexia and allowing the introduction of AI-based educational and clinical interventions on neurodevelopmental disorders. Here, we introduce a hybrid machine learning technique for multimodal dyslexia identification. It will be a combination of the analysis of speech signals with neurophysiological data as well as advanced preprocessing to enhance feature extraction and the quality of data. It utilizes enhanced Support Vector Machines on classifying speech sounds by combining Mel-frequency cepstral coefficients from speech sounds with coherence values from EEG data. This integration has been done to bridge a gap between behavioral symptoms of dyslexia and brain activity for an in-depth knowledge about it.

The system uses Raspberry Pi architecture to make the system both accessible and scalable, therefore ensuring that the processing be real-time while being offered at a

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reasonable cost. This portability makes for broad access and can also be implemented in community health clinics and schools. Furthermore, adaptive reinforcement learning has been used to fine-tune the interventions to the level of dyslexia. By implementing interactional features including audio response and visual feedback while allowing students to play into various game modules, this system makes sure that no learning-related dyslexic issue exists without being faced or covered.

II. Literature Review

Due to the increasing integration of AI within healthcare, advanced diagnostics on diverse cognitive and neurological diseases could be developed. Thus, dyslexia has become a prominent neurodevelopmental disorder affecting reading and writing. For detection and management, tailored ML models enable automated and scalable solutions. Recent studies have explored multimodal data, including speech signals, EEG, and behavioral datasets, to enhance detection accuracy.

The hybrid machine learning application presented in [1] employs an optimized multiclass Support Vector Machine integrated with reinforcement learning for real-time dyslexia detection. The architecture is Raspberry Pi-based, processing speech signals using Mel Frequency Cepstral Coefficients (MFCCs) for feature extraction and classification. It delivers personalized support through an LCD and auditory feedback, enhancing accessibility.

Arns et al. [2] studied EEG power and coherence patterns to reveal distinct brain activation in dyslexic children, validating the double-deficit theory. Increased Delta and Theta activity in specific regions highlight compensatory mechanisms that can inform AI models for improved classification.

The review in [3] examined machine learning approaches for identifying dyslexia biomarkers, addressing challenges such as overfitting, hyperparameter tuning, and data privacy. These insights inform algorithm optimization and robustness.

Further, [4] employed feature optimization techniques like PCA and kernel tuning in SVMs to improve classification accuracy with reduced complexity. A gamified online test structure quantified behavioral and cognitive deficits, complementing traditional assessments.

A classification model discussed in [5] delivered customized tools for university students with dyslexia. It emphasized adaptive strategies and the importance of user-specific interventions through ML frameworks.

Multimodal detection frameworks combining behavioral, neurophysiological, and linguistic data were proposed in [6]. The study supported the use of EEG and speech data to improve diagnostic precision. Evaluations showed the effectiveness of SVM and artificial neural networks (ANNs) for dyslexia classification.

As reviewed in [7], EEG-based classification models can benefit from artifact reconstruction and task-specific data collection. The authors reinforced SVM's applicability to EEG data in diagnosing neurodevelopmental disorders like dyslexia.

The study in [8] applied signal processing and ML to EEG data to identify dyslexia biomarkers. Key issues like noise reduction, entropy analysis, and preservation of clinical record quality were addressed.

In [9], early detection of dyslexia through EEG signal analysis is explored, emphasizing machine learning's role in predicting dyslexia in infants. The study compares classifiers for detecting early patterns in brain activity.

Finally, the work in [10] provides a holistic view of AI's potential in addressing neurocognitive impairments linked to dyslexia. The literature collectively demonstrates how AI and ML innovations, combined with multimodal inputs and feature optimization, are paving the way for scalable and inclusive solutions for early diagnosis and intervention.

III. Existing system

The existing popular tools for dyslexia detection are mainly dependent upon traditional techniques or very basic learning frameworks of machines. The traditionally used methods include professional appraisals, psychoeducational testing and observational assessment. Though beneficial, these are too cost-inefficient and time-consuming that require the availability of professionals which rarely happens in underprivileged resources.

Recently, conventional methods have started to take a backseat, and technology-based methods are starting to come into the fore. Among these, speech analysis is an important method used. Such systems utilize methods like MFCCs and SVM for classification in order to assess the auditory characteristics and find phonological deficiencies in the dyslexic children. These methods, although automated in nature, have drawbacks like regional dialects, background noises, and linguistic variance that could possibly reduce accuracy in real-life applications.

EEG-based techniques also use brain activity patterns, such as delta and theta band activity, to detect dyslexia. Although EEG technology is useful in studying the neurological foundation of dyslexia, clinics and schools with limited resources cannot employ it due to its high cost and complexity.

In addition, while algorithms for optimizing feature extraction such as PCA offer improvements in efficiency, they typically rely on controlled datasets, thus limiting application. Scalability is also addressed by distributed machine learning models but pose problems of difficulty in synchronizing, real-time processing delay, and infrastructure needs.

Despite such advancements, existing systems possess several critical limitations:

1. Accessibility: High costs and reliance on specialized hardware, such as EEG devices, make most of the systems inaccessible to the general population.

2. Scalability: Most solutions are not built to handle large-scale deployment across diverse environments such as schools or rural healthcare facilities.

3. Noise Sensitivity: Speech-based systems are vulnerable to environmental noise, which hinders their performance in real-world applications.

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4. Single-Modality Focus: Most of the systems focus on a single data type, such as speech or EEG, failing to leverage the benefits of multimodal approaches for better accuracy.

5. Ethical Concerns: Obtaining private information, especially from children, may raise ethical concerns around data privacy and legal compliance. Although the existing methods may generally be used as a basis for diagnosing dyslexia, their current drawbacks highlight the need for hybrid frameworks, user-friendly, privacy-conscious systems, and multimodal data that users can access. This is an opportunity to provide robust, expandable, and user-friendly solutions to the problems with the current systems.

IV. Methodology: System Design and Architecture

The Hybrid System Architecture of the developed solution is an advanced machine learning framework, powered by AI, which can be used to detect dyslexia and provide personalized adaptive learning opportunities. Our high-tech architecture, combining multi-modal data analysis, reinforcement learning, and voice signal processing, allows for developing exact and scalable solutions. This solution is superior in terms of overcoming the deficits of the existing models of improved accessibility, efficiency, and accuracy in using cutting-edge machine learning algorithms in conjunction with a robust preprocessing pipeline.

Functional Architecture of the Proposed System:

A. Data Collection: Speech and EEG signals from individuals with and without dyslexia were collected in order to create a balanced sample. A USB microphone was utilized to capture the participants' voice while they read phoneme-rich texts, and non-invasive electrodes were used to collect their EEG data while they read. Among the ethical topics that were covered were informed consent and permission. The preprocessing included artifact removal for EEG, noise reduction, and MFCC-based feature extraction for voice. The dataset was split into training and testing sets, and K-fold cross-validation was applied to enhance model performance and generalization.

B. Training Module: The training module prepares the system by processing benchmark datasets, extracting features, and training machine learning models to classify dyslexia patterns. Dataset Description: This benchmark dataset of speech signals is collected so that it reflects diversity in accents, languages, and speech patterns. It has been expanded for multilingual and accented speech samples to cover linguistic variability; thus, the system could be globally applied.

Advanced Preprocessing:

• Noise Reduction: The background noise removal is obtained by using different adaptive noise reduction techniques; these are Wiener filtering, and the spectral subtraction techniques.
• Normalization: To normalize speech sounds to a specific amplitude, resampling techniques are used. The signal that has been normalized is depicted as

$$S_{\text{norm}}(t) = \frac{S(t)}{\max|S(t)|} \tag{1}$$

Where $S(t)$ represents a speech signal input, while $S_{\text{norm}}(t)$ is the normalized signal.

Key Attribute Identification:

DFT defines the frequency components of the speech signal and yields Mel Frequency Cepstral Coefficients (MFCC):

$$\text{MFCC}(k) = \log\left(\sum_{n=1}^{N}|X(n)|^2 H_m(f(n))\right) \tag{2}$$

Where $X(n)$ is the DFT of the signal, H_m is the triangular filter, and $f(n)$ is the frequency.

Wavelet Transform : Complement to MFCC, wavelet transforms analyze time andfrequency information in an enhanced way.

Classification:

Support Vector Machine (SVM): The ensemble consists of classification-based kernel-optimized SVMs. The decision boundary, therefore, is established upon margin optimization between the classes and those that separate them.

$$LP = \frac{1}{2}\|\omega\|^2 + C\sum_{i=1}^{N}\varepsilon_i \tag{3}$$

Subject to: $y_i(\omega \cdot x_i + b) \geq 1 - \varepsilon_i,\ \varepsilon_i \geq 0$

Here,The mass of the vectors is denoted by ω ,the deviation by b , the surplus variable by ε_i , and C is the penalty parameter.

C. Testing Module: This module tests real-time speech signals to evaluate the learned model, identify dyslexia, and give differential learning experiences.

1) Data Gathering:

A USB microphone will record speech signals in real time during reading activities. Pre-processing of the audio will occur within the Raspberry Pi architecture to improve its quality.

Directional Filtering: This process utilizes directional microphones and adaptive filters to filter out background noises and focus on the voice of the speaker.

Classification: The trained SVM module is fed with real-time speech data to classify patterns of dyslexia. The system calculates a probability score of dyslexia based on feature weights.

2) Adaptive Reinforcement Learning: :

The child's performance is used as input to the system to vary its response. For example, if difficulties are detected by the system, it simplifies the tasks for the child but increases the complexity as its performance improves.

V. EXPERIMENTAL SETUP

In this section, the experimental setup of the proposed Raspberry Pi-based application for dyslexia detection is elaborated, incorporating essential hardware modules, configurations, and additional features to enhance system performance. The experimental setup ensures seamless

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functionality in real-time speech processing, dyslexia detection, and personalized learning feedback.

The experimental configuration of the Raspberry Pi-based dyslexia detection program is described in detail in this section. To improve system performance, it integrates necessary hardware modules, settings, and features. Fig. 1 displays the entire setup, guaranteeing smooth operation of dyslexia detection, real-time speech processing, and customized learning feedback.

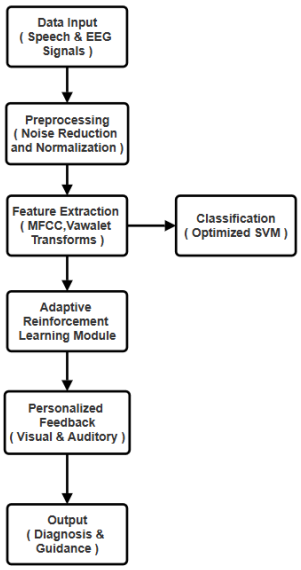


Fig. 1: Block diagram of the experimental setup

A. USB microphone The USB plug-and-play desktop microphone is used for picking up speech signals from the children during real-time experiments. Its high sensitivity coupled with noise rejection ability leads to clear and quality input audio, which is considered essential for accurate processing as well as feature extraction. Additionally, the microphone is constructed in light metal and is compatible with the Raspberry Pi, making it portable.

B. The Raspberry Pi: The central processing unit in this program is the Raspberry Pi 4 Model B+. It includes a 64-bit, quad-core CPU with high-speed, real-time calculation capabilities that runs at 1.5 GHz.
Wireless: double-band Wi-Fi along with Bluetooth capabilities provides efficient data transfer and an option to integrate with a cloud service. Ports and GPIO Heads: Equipped with HDMI for external display connection and 40-pin GPIO Heads for expansions and further additional peripherals. Scalability: It supports light weight machine learning tasks and it interfaces with other hardware for modular extensions.

C. LCD Screen: An on-board 5-inch LCD monitor is included for visualization feed. It is meant for real-time display of dyslexia likelihood percentage as percentage. Instructions for fostering tools such as personalized learning tasks or reinforcement activities.

D. Speaker Module: An external speaker module is added to deliver auditory feedback and reinforcement instructions. This is particularly helpful for children with dyslexia, as auditory cues can significantly improve phonological and reading skills.

E. Battery Backup System: To make it portable and reliable, this system is integrated with a rechargeable battery pack, ensuring that it can work seamlessly in classrooms, clinics, and remote locations. The system has been designed to back up the Raspberry Pi, LCD display, USB microphone, and speaker module for as long as possible.

Algorithm

Input:SpeechSignal

Pre-Processing

- **Noise Reduction** _ Median Filter
  Background Removal (Speech Signal)
  Undesirable Noise Elimination (Speech Signal)

Normalization ()
  Resampling (Speech Signal)
  Amplitude of Speech Signal Adjusted to a Fixed Rate
  Feature Extraction _ Fast Fourier Transform
  MFCC Features = Pitch, Contrast, Centroid, RMS

Feature Extraction_Support Vector Machine
  Vector V[]: Represents MFCC Features
  Distribute Vector in a Plane
  Compute Hyperplane _ Decision Boundary
  Feature Margin () Calculation
  Compute Feature Weight using Lagrangian Multiplier
  Feature Weight (FwF _ wFw) = Feature Score (Likelihood Percentage)
  Classification (Likelihood Percentage)

Output = {Dyslexia, Non-Dyslexia}
if Dyslexia Likelihood Percentage ¿ Threshold **then**
  Enable Fostering Personalized Learning Tool
else
  if Dyslexia Likelihood Percentage ¡= Threshold **then**
    Deliver Adaptive Feedback via Speaker Module
  end if
  Display Dyslexia Likelihood Percentage on LCD Display
end if

VI. EXPERIMENTAL ANALYSIS

The system is tested with k-fold cross-validation to ensure strong testing on several subsets of the dataset, decrease overfitting, and give a fair judgment of how good the model is performing.Confusion matrix measures accuracy by how well classifier predictions compare against actual labels. FP , FN , TP , TN are key measures. It uses these parameters to compute performance measures that ensure effective categorization. This part deals with the experimental evaluation of the proposed Raspberry Pi-based dyslexia detection model. It analyses the children's speech signals by identifying the

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patterns of dyslexia using SVM and extracts the characteristics of MFCC. The integration of Python with Raspberry Pi allows accurate dyslexia detection and real-time processing. The LCD display proportion of people who have dyslexia after processing voice input from a USB microphone in real time. Powerful libraries, SVM classification, feature extraction and signal preprocessing are all by Python. The LCD Display of findings of dyslexia detection is shown at the hardware level in Fig. 1. Includes accuracy, precision, recall, and F1 score.

VII. THE PERFORMANCE EVALUATION

Cross-validation and a confusion matrix guarantee the F1 rating, recall, accuracy, and precision. of the dyslexia detection system. It correctly classifies speech signals as either dyslexic or not. The study highlights how well a Python-based solution and Raspberry Pi hardware work in the process of machine learning.

Cross-Validation and Confusion Matrix: To guarantee that the system is properly tested on different dataset subsets, overfitting is prevented, and the model offers an objective assessment of its performance, it is trained using k-fold cross-validation. The confusion matrix shown in Fig. 2 calculates accuracy by comparing classifier predictions to actual labels. FP, FN, TP, and TN are all significant metrics. By using these parameters to calculate performance metrics like accuracy, precision,sensitivity and F1 rating, proper categorization is maintained.

Cross-validation and a confusion matrix guarantee the F1 rating, recall, accuracy, and precision of the dyslexia detection system. It correctly classifies speech signals as either dyslexic or not. The study highlights how well a Python-based solution and Raspberry Pi hardware work in the process of machine learning.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

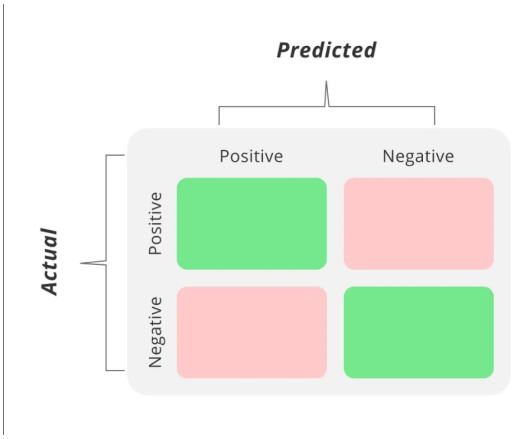


Fig. 2: Confusion matrix used for classification evaluation

Precision: Precision is the fraction of correctly classified dyslexics out of all classifications or cases classified as dyslexia.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

Recall or sensitivity: the ratio of number of actual dyslexic cases actually diagnosed.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

F1 Rating:

$$\text{F1 Score} = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \tag{7}$$

F1 Rating will balance the trade-off between false positives and negatives through an arithmetic mean of accuracy and recall.

Precision and Efficiency: The number of successfully identified samples divided by the total number of samples may be used to determine the overall accuracy of the model.

The comparative performance of the models in terms of precision, recall, and F1 score is illustrated in Fig. 3.

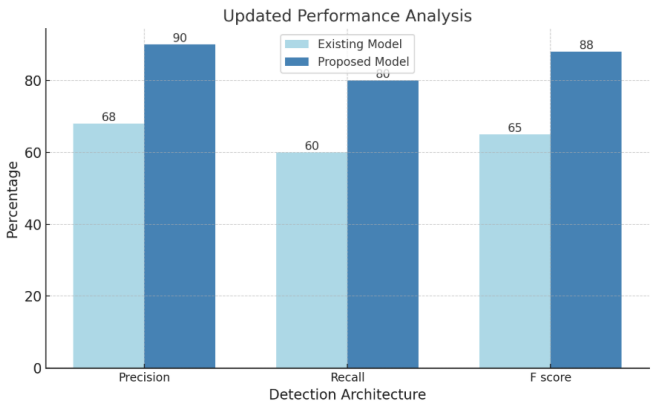


Fig. 3: Performance analysis of Model A vs Model B

The corresponding performance metrics are summarized in Table I.

Metric	Model A (%)	Model B (%)
Precision	70	88
Recall	62	82
F1 Score	67	90

TABLE I: Performance evaluation of classification models

Accurate: It is evident that the suggested model uses effective classification with SVM and robust feature extraction with MFCC to provide higher accuracy. These features guarantee that the algorithm can distinguish between a few modest phonological patterns found in the speech of dyslexics.

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VIII. Future Plan

Future studies will concentrate on increasing classifier performance, incorporating deep learning models, and polishing feature extraction in order to improve multimodal dyslexia detection. Model generalization will be improved by addressing phoneme-level analysis, language variability, and EEG standardization. Gamification and AI-driven task modifications will be added to adaptive reinforcement learning for tailored treatments. Edge AI, cloud integration, and practical validation in clinics and schools will guarantee scalability. Deployment will be guided by ethical issues, such as privacy-preserving AI and regulatory compliance. AI-driven dyslexia screening will be supported by cooperation with educators and legislators, guaranteeing inclusive diagnostic and intervention options, standardization, and accessibility. This AI-powered dyslexia detection system on a Raspberry Pi is used by a teacher who observes a student having difficulty with reading fluency. An EEG headgear records brain activity as the learner reads aloud, and a USB microphone records speech to analyze fluency and articulation. Data is preprocessed by the system, which extracts coherence measures for EEG and MFCCs for speech. After analyzing patterns, a trained SVM classifier produces a dyslexia likelihood score. Personalized phonics activities are offered by adaptive reinforcement learning if it is high. Learning is aided by immediate visual and aural feedback through speakers and an LCD. Early dyslexia assistance in schools is guaranteed by this low-cost, real-time approach.

IX. Conclusion

This paper introduces a hybrid framework for real-time dyslexia diagnosis using Raspberry Pi hardware and Python programming. In the system, preprocessing is included such as noise reduction and normalization, then extraction of features using MFCC, and classification using an improved Support Vector Machine. There's the speaker module, offering adaptive aural feedback and reinforcement tasks based on individual probability scores. This allows every student to receive individualized instruction with further continuous improvement of their skills. The system, based on experimental data, correctly identifies patterns related to dyslexia with high accuracy, precision, and recall. Because it is affordable and scalable, it may be used in settings with limited resources, such as clinics and schools. Interactive and gamified learning materials improve usability and engagement even further. This framework provides a dependable, scalable, and user-friendly solution, which is the major advancement over traditional methods. Its worldwide applicability will increase with future improvements like cloud analytics and multilingual dataset support.

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